

Notes on Mathematical Logic

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Notes based on the Mathematical Logic lecture series by Professor Artem Chernikov: <https://tinyurl.com/MathLogicLectureSeries>

Introduction

As mentioned in the title page, this document is a set of notes based on the Mathematical Logic lecture series by Professor Artem Chernikov, which can be found on YouTube (<https://tinyurl.com/MathLogicLectureSeries>). The video series covers topics from the Mathematical Logic course (Math 220A) taught by Artem Chernikov at UCLA in the Fall quarter of 2021. The lectures themselves follow Chapters 2 and 3 of "A First Journey Through Logic" by Martin Hils and François Loeser and later discusses further material in model theory from David Marker's "Model Theory : An Introduction".

At this particular moment I am unsure how much I will expand on these notes beyond the content of Professor Artem's course. As a disclaimer, these notes are written by a novice and have no affiliation with the actual course or Professor Artem. It would be highly appreciated if any errors were brought to my attention.

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Chapter 1

First-Order Logic

1.1 Languages and Structures

The main idea of this section is to introduce the concept of languages and structures with regards to first-order logic. The gist of it is as follows:

A **formal language** consists of variables, terms, formulas, and other symbolic constructions. It also consists of formal deduction rules, i.e. what allows us to deduce statements from axioms via formal proofs. The main point of a formal language is that it is **syntactic** in nature. That is, the symbols we use have no inherent meaning.

In contrast, **structures** are sets (or models) where it is possible to "interpret" terms and formulas. That is, one may give meaning to the validity/satisfaction of a formula in a formal language (with no free variables) inside a given structure. The main point of a structure is that it is **semantic** in nature, and provides meaning to the symbols we use.

The main result in this chapter is Gödel's Completeness Theorem which states that a formula with no free variables can be formally deduced from a given set of axioms if and only if it is valid in every structure satisfying these axioms. This result essentially gives an equivalence between the syntactic and semantic viewpoints. It also provides an *a posteriori* justification for our choice of syntax.

In first-order logic (FOL), statements are just sequences of symbols which describe properties of certain mathematical structures. Consider, for example, the expression

$$\forall x (x > 0 \implies \exists y \ y \cdot y = x).$$

This is a sequence of symbols with no meaning attached *a priori*. However, in a mathematical interpretation, it is a sentence which holds in an ordered field if and only if any positive element is a square. In particular, it holds in the ordered field of real numbers $\langle \mathbb{R}; 0, 1, +, -, \cdot, < \rangle$, but does not hold in the ordered field of rational numbers $\langle \mathbb{Q}; 0, 1, +, -, \cdot, < \rangle$ as the rational number 2 has no "rational root" satisfying the sentence.

Remark 1.1.1

It is important to note, however, that some other properties of ordered fields (e.g. completeness, Archimedean property) cannot possibly be expressed in FOL.

Now, in order for us to discuss all of these notions in a rigorous manner, we must formally define a language.

Definition 1.1.2: First-Order Language

A (first-order) language is a set of symbols $\mathcal{L}$ composed of two disjoint subsets:

1. The first part (which is common to all languages) consists of “(” and “)” together with the following logical symbols:
 - The set of variables: $\mathcal{V} = \{v_n \mid n \in \mathbb{N}\}$.
 - The equality symbol: “=”.
 - Connectives: “ $\neg$ ” (negation), “ $\wedge$ ” (conjunction).
 - The existential quantifier: “ $\exists$ ”.

which are just elements of some abstract set $\mathcal{L}$, to which no meaning is attached.

2. The second part (which is specific to $\mathcal{L}$), called the signature of $\mathcal{L}$, denoted $\sigma^{\mathcal{L}}$, consists of the non-logical symbols of $\mathcal{L}$. It consists of the following:
 - A set of constant symbols: $\mathcal{C}^{\mathcal{L}}$.
 - A sequence of sets: $\mathcal{F}_n^{\mathcal{L}}, n \in \mathbb{N}^*$, where elements of $\mathcal{F}_n^{\mathcal{L}}$ are called the n -ary function symbols.
 - A sequence of sets: $\mathcal{R}_n^{\mathcal{L}}, n \in \mathbb{N}^*$, where elements of $\mathcal{R}_n^{\mathcal{L}}$ are called the n -ary relation symbols (or n -ary predicates).

The language $\mathcal{L}$ is given by the (disjoint) union of these sets.

We remark that although the names of the subsets of a language are rather indicative of some possible interpretation, we have (at this point) yet to assign any meaning to them whatsoever.

Remark 1.1.3

A language is always infinite. We note that the logical part of a language is countable, however there is no restriction on the cardinality of the set of non-logical symbols (the signature) of a language.

Remark 1.1.4

Since the logical part of a language is common to all languages, we introduce an abuse of notation. That is, we often identify a language $\mathcal{L}$ by its signature $\sigma^{\mathcal{L}}$.

E.g. Examples of languages:

1. The language

$$\mathcal{L}_\emptyset = \emptyset$$

is the empty language which contains no symbols.

2. The language

$$\mathcal{L}_{\text{ring}} = \{0, 1, +, -, \cdot\}$$

is the ring language where

- 0 and 1 are constants in $\mathcal{C}^{\mathcal{L}_{\text{ring}}}$, respectively interpreted as the additive and multiplicative identities;
- $-$ is a unary function in $\mathcal{F}_1^{\mathcal{L}_{\text{ring}}}$, interpreted as the additive inverse of an element; and
- $+$ and $\cdot$ are binary functions in $\mathcal{F}_2^{\mathcal{L}_{\text{ring}}}$, respectively interpreted as the usual ring addition and multiplication.

3. The language

$$\mathcal{L}_{\text{ord}} = \{<\}$$

is the order language where $<$ is a binary relation in $\mathcal{R}_2^{\mathcal{L}_{\text{ord}}}$.

4. The language

$$\mathcal{L}_{\text{o.ring}} := \mathcal{L}_{\text{ring}} \cup \mathcal{L}_{\text{ord}}$$

is the ordered ring language.

5. The language

$$\mathcal{L}_{\text{set}} = \{\in\}$$

is the language of set theory, where $\in$ is a binary relation in $\mathcal{R}_2^{\mathcal{L}_{\text{set}}}$.

6. The language

$$\mathcal{L}_{\text{grp}} = \{\cdot, ^{-1}, 1\}$$

is the language of groups where

- $\cdot$ is a binary function in $\mathcal{F}_2^{\mathcal{L}_{\text{grp}}}$, interpreted as the binary group operation;
- $^{-1}$ is a unary function in $\mathcal{F}_1^{\mathcal{L}_{\text{grp}}}$, interpreted as the inverse function; and
- 1 is a constant in $\mathcal{C}^{\mathcal{L}_{\text{grp}}}$, interpreted as the identity element.

7. The language

$$\mathcal{L}_{\text{graphs}} = \{E\}$$

is the language of graphs where E is a binary relation in $\mathcal{R}_2^{\mathcal{L}_{\text{graphs}}}$, interpreted as the edge relation.

8. The language

$$\mathcal{L}_{\text{ar}} = \{0, S, +, \cdot, <\}$$

is the language of arithmetic where

- 0 is a constant in $\mathcal{C}^{\mathcal{L}_{\text{ar}}}$, interpreted as zero;

- S is a unary function in $\mathcal{F}_1^{\mathcal{L}_{\text{ar}}}$, interpreted as the successor function;
- $+$ and $\cdot$ are binary functions in $\mathcal{F}_2^{\mathcal{L}_{\text{ar}}}$, respectively interpreted as addition and multiplication; and
- $<$ is a binary relation in $\mathcal{R}_2^{\mathcal{L}_{\text{ar}}}$, interpreted as ordering.

It is important to note here that the text in blue which gives meaning and a way to interpret the symbols is NOT inherent to the first-order language. Recall that a formal language is syntactic, and devoid of inherent meaning. The above languages are, at this moment, arbitrary sets of symbols and nothing more. If we wish to actually assign meaning in the way that the colored text does, we must discuss structures.

Definition 1.1.5: $\mathcal{L}$ -structures

Let $\mathcal{L}$ be a first-order language. An $\mathcal{L}$ -structure, $\mathcal{A}$, consists of:

- a nonempty set A , called the base set of $\mathcal{A}$;
- an element $c^{\mathcal{A}} \in A$ for each $c \in C^{\mathcal{L}}$;
- a function $f^{\mathcal{A}} : A^n \rightarrow A$ for each $f \in \mathcal{F}_n^{\mathcal{L}}$; and
- a subset $R^{\mathcal{A}} \subseteq A^n$ for each $R \in \mathcal{R}_n^{\mathcal{L}}$.

We write $\mathcal{A} = \langle A; (Z^{\mathcal{A}})_{Z \in \sigma^{\mathcal{L}}} \rangle$ where $Z^{\mathcal{A}}$ is called the interpretation of the symbol $Z \in \sigma^{\mathcal{L}}$ in the structure $\mathcal{A}$.

E.g. Examples of $\mathcal{L}$ -structures:

1. Consider $\mathcal{N} = \langle \mathbb{N}; 0, S, +, \cdot, < \rangle$ to be the structure with the following interpretations:
 - The base set $\mathbb{N}$ is the set of natural numbers.
 - The constant symbol 0 , interpreted as the actual number 0 .
 - The unary function symbol S , interpreted as the successor function on $\mathbb{N}$ ($x \mapsto x + 1$).
 - The binary function symbols $+$ and $\cdot$, respectively interpreted as the usual addition and multiplication on the natural numbers.
 - The binary relation symbol $<$, interpreted as the usual ordering on the natural numbers.

Then, $\mathcal{N}$ is an $\mathcal{L}_{\text{ar}}$ -structure.

2. Consider $\mathcal{C} = \langle \mathbb{C}; 0, 1, +, -, \cdot \rangle$ to be the structure with the following interpretations:
 - The base set $\mathbb{C}$ is the set of complex numbers.
 - The constant symbols 0 and 1 , respectively interpreted as the complex numbers 0 and 1 .
 - The unary function symbol $-$, interpreted as the additive inverse function ($x \mapsto -x$).
 - The binary function symbols $+$ and $\cdot$, respectively interpreted as the usual addition and multiplication on the complex numbers.

Then, $\mathcal{C}$ is an $\mathcal{L}_{\text{ring}}$ -structure. Specifically, it is the field of complex numbers.

3. Consider $R = \langle \mathbb{R}; 0, 1, +, -, \cdot, < \rangle$ to be the structure with the following interpretations:

- The base set $\mathbb{R}$ is the set of real numbers.
- The constant symbols 0 and 1, respectively interpreted as the real numbers 0 and 1.
- The unary function symbol $-$, interpreted as the additive inverse function ($x \mapsto -x$).
- The binary function symbols $+$ and $\cdot$, respectively interpreted as the usual addition and multiplication on the real numbers.
- The binary relation symbol $<$, interpreted as the usual ordering on the real numbers.

Then, R is an $\mathcal{L}_{\text{o.ring}}$ -structure. Specifically, it is the ordered field of real numbers.